FreeBuff v4

Multi-Modal Brain Tumor Classification & Clinical Decision Support

Diagnostic Imaging Classification Report

Generated By	FreeBuff v4 Research System
Report Date/Time	2026-09-13 15:24:06
Modality	MRI Brain
Scan Description	Brain MRI slice — classification target: Pituitary
Patient ID	TEST-001
Exam Date	2026-09-13

CLASSIFICATION RESULTS

Predicted Class: Pituitary

Prediction Confidence: 0.9532 (95.32%) — Confidence Band: High

Predicted Class Index: 3

Test-Time Augmentation (TTA) Used: True

Source Image: dashboard upload

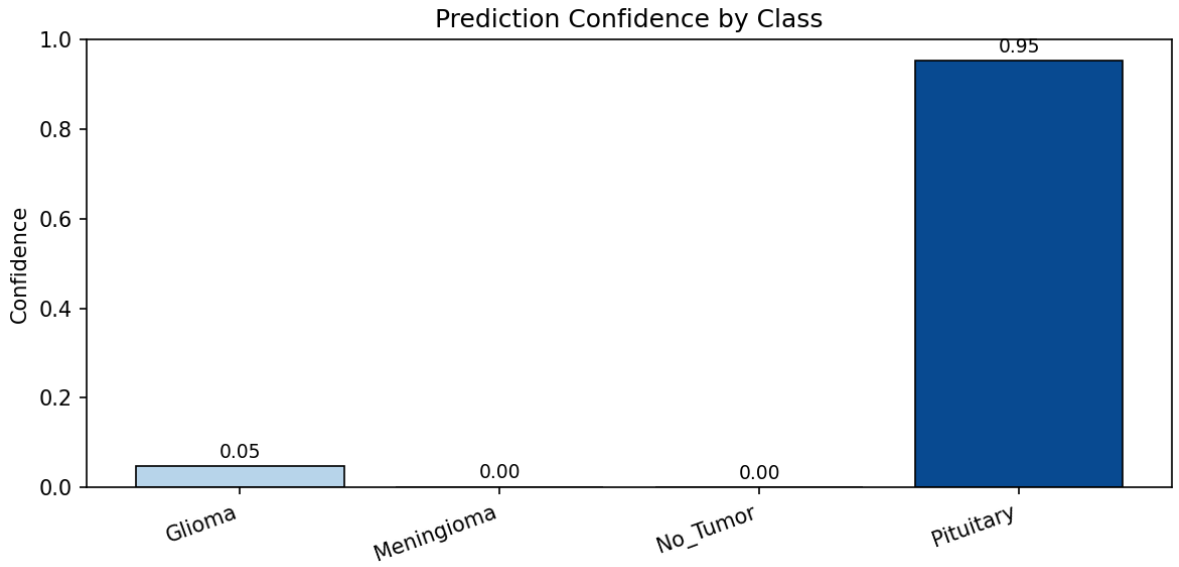
Per-Class Probability Distribution

Class	Probability	Percentage
Glioma	0.0463	4.63%
Meningioma	0.0005	0.05%
No_Tumor	0.0000	0.00%
Pituitary	0.9532	95.32%

Raw Logits (Pre-Softmax)

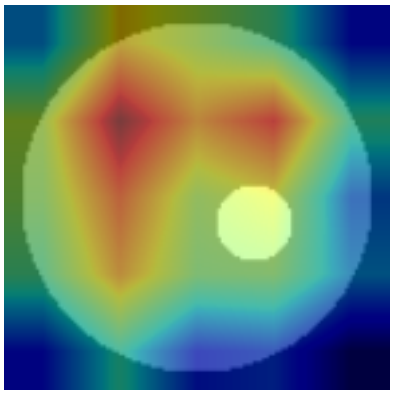
Class	Logit
Glioma	0.0000
Meningioma	0.0000
No_Tumor	0.0000
Pituitary	0.0000

CONFIDENCE SPECTRUM VISUALIZATION



The bar chart above displays the predicted probability (confidence) assigned to each tumour class by the ensemble decision pathway. Higher bars indicate stronger model assertion for the corresponding diagnostic category.

GRAD-CAM++ EXPLAINABILITY HEATMAP



The heatmap above visualizes the image regions most influential in the model's classification decision (Grad-CAM++). Warmer colors indicate higher importance. This visualization is intended to assist clinician review and is NOT a standalone diagnostic indicator.

WHO GRADE & STAGING ASSESSMENT

Tumour Class: **Pituitary**

Grade Label: WHO Grade I (Benign Non-Functioning / Small Adenoma)

Typical Imaging Features: Well-circumscribed sellar/suprasellar lesion, homogeneous or mildly heterogeneous enhancement, possible microadenoma <10 mm, minimal or no cavernous sinus invasion.

Malignancy Profile: Benign pituitary adenoma; rarely malignant. Hormone secretion status drives management more than grade.

Growth Pattern: Slow, expansile, may compress optic chiasm.

Recommended Intervention Protocol: Endocrine evaluation and hormone panel; observation for incidental asymptomatic microadenomas; surgical resection (transsphenoidal) for visual compromise or

hormone-secreting tumours; medical therapy (dopamine agonists for prolactinomas); radiation reserved for refractory cases.

CLINICAL REASONING NARRATIVE

FINDINGS Predicted class : Pituitary Prediction confidence: 95.3% Reliability band : High

CLASS PROBABILITY SPECTRUM Glioma 4.63% # Meningioma 0.05% No_Tumor 0.00% Pituitary 95.32% ##############################

WHO STAGING REFERENCE Grade : WHO Grade I (Benign Non-Functioning / Small Adenoma) Imaging features : Well-circumscribed sellar/suprasellar lesion, homogeneous or mildly heterogeneous enhancement, possible microadenoma <10 mm, minimal or no cavernous sinus invasion. Malignancy : Benign pituitary adenoma; rarely malignant. Hormone secretion status drives management more than grade. Growth pattern : Slow, expansile, may compress optic chiasm. Recommended action : Endocrine evaluation and hormone panel; observation for incidental asymptomatic microadenomas; surgical resection (transsphenoidal) for visual compromise or hormone-secreting tumours; medical therapy (dopamine agonists for prolactinomas); radiation reserved for refractory cases.

Source image: in-memory upload

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